



FINAL MARK

**GIRRAWEEEN HIGH SCHOOL
MATHEMATICS EXTENSION 1
HSC ASSESSMENT TASK 2 2007
ANSWERS COVER SHEET**

Name: _____

QUESTION	MARK	HE1	HE2	HE3	HE4	HE5	HE6	HE7
Q1	/15	✓						✓
Q2	/23	✓						✓
Q3	/10	✓						✓
Q4 a,b	/7	✓						✓
Q4c	/5	✓	✓					✓
Total Q4	/12							
Q5	/23	✓						✓
Q6a	/6	✓		✓				✓
Q6b,c	/14	✓						✓
Total Q6	/20							
TOTAL								
	/ 103	/103	/5	/6				/103



GIRRAWEE HIGH SCHOOL

YEAR 12 HALF YEARLY EXAMINATION

Task 2

2007

MATHEMATICS

Extension 1

*Time allowed – Two hours
(Plus 5 minutes reading time)*

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question.
- Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Start each question on a new page.
- A table of standard integrals is provided.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1 (15 marks)

(a) Find $\lim_{x \rightarrow 0} \frac{\sin \frac{x}{3}}{x}$ (Show working) 2

(b) Solve: $\frac{3x-2}{5x-4} + 1 < 0$ 3

(c) Find the acute angle between the lines $3x - 4y = 3$ and $x - 2y = 11$. 4

(d) Sketch: (i) $y = 3 \cos 2x$, $-\pi \leq x \leq \pi$ 3

(ii) $y = \sin \frac{x}{2}$, $-2\pi \leq x \leq 2\pi$ 3

Question 2 (23 marks)

(a) Differentiate: 17

(i) $y = e^{\ln x}$ (ii) $y = \log \sin(x^3)$

(iii) $y = x^2 \tan 2x$ (iv) $y = \frac{\cos x}{x}$

(v) $y = e^{\sin 5x}$ (vi) $y = \log_e \left(\frac{x-5}{x+5} \right)$

(b) Find the equation of the normal to the curve $y = \log_e x$ at the point where $x = 1$. 3

(c) The line $y = mx$ is tangent to the curve $y = e^{4x}$. Find the coordinates of the point of contact. 3

Question 5 (23 marks)

(a) Find:

14

(i) $\int \frac{x+4}{x^2+8x-5} dx$

(ii) $\int (\frac{1}{2} \sin 2x - \cos x) dx$

(iii) $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \sin^2 x dx$

(iv) $\int_0^{\frac{\pi}{6}} \sin x \cos x dx$

(b) The region under the curve $y = \tan x$ between $x = \frac{\pi}{4}$ and $x = \frac{\pi}{3}$ is rotated about the x axis. Find the volume of the solid of revolution.

4

(c) (i) Differentiate $y = \cos^3 x$

2

(ii) Hence evaluate $\int_0^{\frac{\pi}{2}} \cos^2 x \sin x dx$.

3

Question 6 (20 marks)

(a) It is known that 4% of electric bulbs produced by a certain factory are defective. What is the probability that from a random sample of 20 bulbs

(i) all are good.

2

(ii) exactly two bulbs are defective.

2

(iii) at least one bulb is defective.

2

(b) solve:

(i) $\sin 2x = -\frac{\sqrt{3}}{2}, \quad 0 \leq x \leq 360^\circ$

4

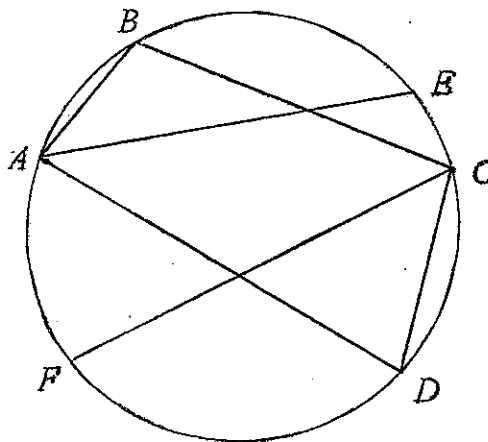
(ii) $3 \cos x + 4 \sin x = 4$ using t method.

5

(c) In the diagram given, $ABCD$ is a cyclic quadrilateral where AE bisects $\angle BAD$ and CF bisects $\angle BCD$. Prove that EF is a diameter.

5

(Hint: Draw $\overline{AF}$ and $\overline{EF}$)



Year 12 Half Yearly 2007 - Extension 1 - Solutions

Question 1 (15 marks)

(a) $\lim_{x \rightarrow 0} \frac{\sin \frac{2x}{3}}{\frac{2x}{3}}$

$$= \lim_{x \rightarrow 0} \frac{\sin \frac{2x}{3}}{3 \times \frac{2x}{3}}$$

$$= \frac{1}{3} \lim_{x \rightarrow 0} \frac{\sin \frac{2x}{3}}{\frac{2x}{3}} \quad (2)$$

$$= \frac{1}{3}$$

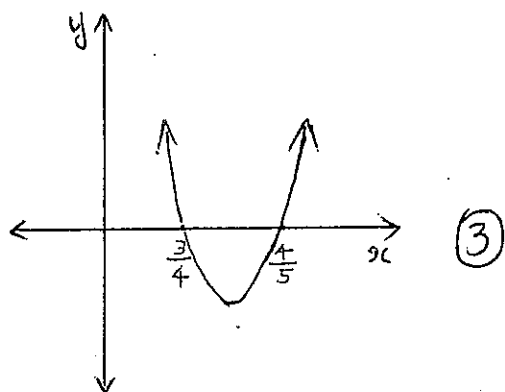
(b) $\frac{3x-2}{5x-4} + 1 < 0$

$$\frac{(5x-4)^2 \times \frac{3x-2}{5x-4} + (5x-4)^2}{5x-4} < 0$$

$$(5x-4)(3x-2) + (5x-4)^2 < 0$$

$$(5x-4)(3x-2+5x-4) < 0$$

$$(5x-4)(8x-6) < 0$$



From the graph the solution is

$$\frac{3}{4} < x < \frac{4}{5}$$

(c) $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$$3x - 4y = 3$$

$$x - 2y = 11$$

$$3x - 3 = 4y$$

$$x - 11 = 2y$$

$$y = \frac{3}{4}x - \frac{3}{4}$$

$$y = \frac{1}{2}x - \frac{11}{2}$$

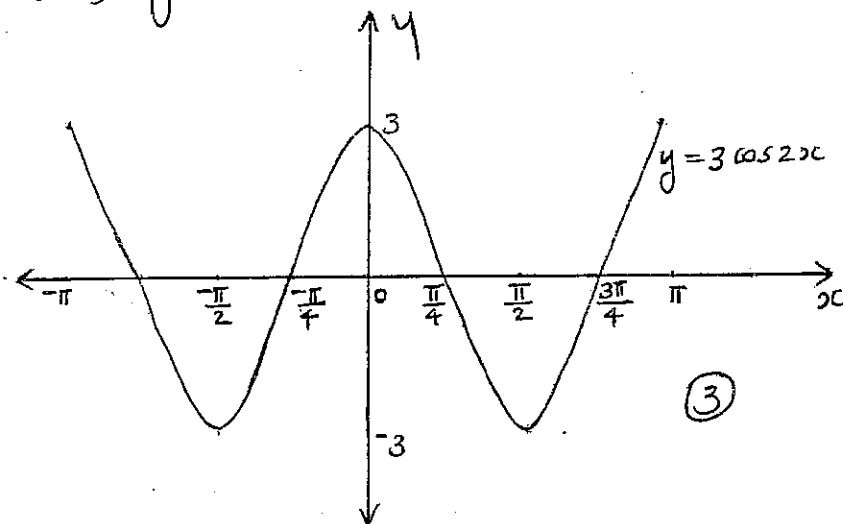
$$m_1 = \frac{3}{4}$$

$$m_2 = \frac{1}{2}$$

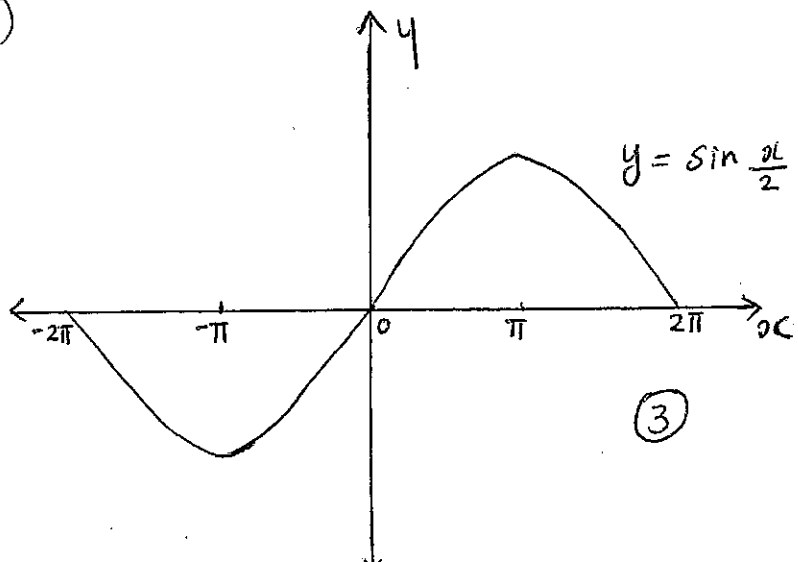
$$\tan \theta = \left| \frac{\frac{3}{4} - \frac{1}{2}}{1 + \frac{3}{4} \times \frac{1}{2}} \right| = \frac{2}{11} \quad (4)$$

$$\theta = 10^\circ 18'$$

(d) (i) $y = 3 \cos 2x$, $-\pi \leq x \leq \pi$



(ii)



Question 2 (23 marks)

(a) (i) $y = e^{\ln x}$
 $= x$

$$\frac{dy}{dx} = 1 \quad (2)$$

(ii) $y = \log \sin(x^3)$

$$\begin{aligned} y' &= \frac{1}{\sin(x^3)} \times \cos(x^3) \times 3x^2 \\ &= \frac{3x^2 \cos(x^3)}{\sin(x^3)} \\ &= \underline{\underline{3x^2 \cot(x^3)}} \quad (3) \end{aligned}$$

(iii) $y = x^2 \tan 2x$

$$\begin{aligned} y' &= x^2 \times (\sec^2 2x) \times 2 + \tan 2x \times 2x \\ &= \underline{\underline{2x^2 \sec^2 2x + 2x \tan 2x}} \quad (3) \end{aligned}$$

(iv) $y = \frac{\cos x}{x}$

$$\begin{aligned} y' &= \frac{x \times -\sin x - \cos x \times 1}{x^2} \\ &= \underline{\underline{\frac{-x \sin x - \cos x}{x^2}}} \quad (3) \end{aligned}$$

(v) $y = e^{\sin 5x}$

$$\begin{aligned} y' &= e^{\sin 5x} \times \cos 5x \times 5 \\ &= \underline{\underline{5 \cos 5x e^{\sin 5x}}} \quad (3) \end{aligned}$$

(vi) $y = \log_e \left(\frac{x-5}{x+5} \right)$
 $= \log_e(x-5) - \log_e(x+5)$

$$y' = \frac{1}{x-5} - \frac{1}{x+5} \quad (3)$$

(b) $y = \log_e x$

$$\frac{dy}{dx} = \frac{1}{x}$$

when $x=1$, $\frac{dy}{dx} = 1$

gradient of the normal = -1

when $x=1$, $y = \log_e 1 = 0$

gradient -1 and the point is (1,0)

Equation of the normal is

$$y - 0 = -1(x - 1) \quad (3)$$

$$\underline{\underline{y = -x + 1}}$$

(c) $y = mx$ — (1) $y = e^{4x}$ — (2)

At the point of contact of (1) and (2) we have $mx = e^{4x}$

$$m = \frac{e^{4x}}{x}$$

Gradient of the curve $y = e^{4x}$ at the point of contact is equal to the Gradient of $y = mx$

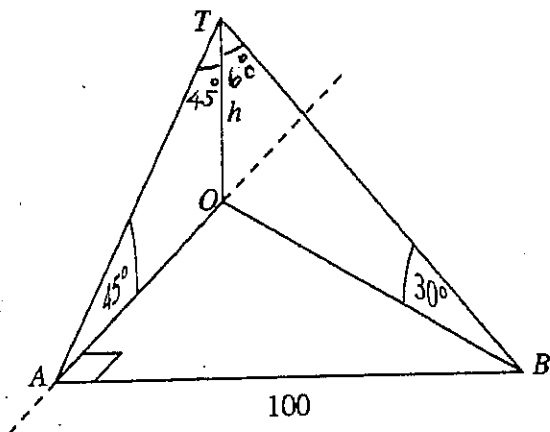
$$4e^{4x} = m$$

$$4e^{4x} = \frac{e^{4x}}{x} \quad \therefore x = \frac{1}{4}$$

when $x = \frac{1}{4}$, $y = e^{4 \times \frac{1}{4}} = e \quad (3)$

The point of contact is $\underline{\underline{\left(\frac{1}{4}, e \right)}}$

Question 3 (10 marks)



$$(i) \tan 60 = \frac{OB}{h} \quad (2)$$

$$\sqrt{3} = \frac{OB}{h} \quad \therefore OB = \sqrt{3}h$$

$$(ii) \tan 45 = \frac{OA}{h}$$

$$1 = \frac{OA}{h} \quad \therefore OA = h$$

By Pythagoras' theorem in $\triangle OAB$

$$OB^2 = OA^2 + 100^2$$

$$(\sqrt{3}h)^2 = h^2 + 100^2$$

$$3h^2 = h^2 + 100^2 \quad (2)$$

$$2h^2 = 100^2$$

$$h = \frac{100}{\sqrt{2}} = 50\sqrt{2}$$

$$(iii) \tan \angle AOB = \frac{100}{50\sqrt{2}} = \sqrt{2}$$

$$\angle AOB = 55^\circ$$

The bearing of B from the base of the tower is $180 - 55 = 125^\circ$ (2)

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(b) (i) PROBLEM - 7 letters

$$\begin{matrix} P \\ \bigcirc \\ 1 \end{matrix} \quad \begin{matrix} \bigcirc \\ 6 \end{matrix} \quad \begin{matrix} \bigcirc \\ 5 \end{matrix} \quad \begin{matrix} \bigcirc \\ 4 \end{matrix}$$

$$\begin{aligned} \text{Total number of arrangements} \\ &= 1 \times 6 \times 5 \times 4 \quad (2) \\ &= 120 \end{aligned}$$

$$(ii) \quad \bigcirc \quad \bigcirc \quad \bigcirc \quad \bigcirc$$

B can be arranged in 4 ways and the remainder in $5 \times 4 \times 3$ ways

$$\begin{aligned} \text{Total number of arrangements} \\ &= 4 \times 5 \times 4 \times 3 \quad (2) \\ &= 240 \end{aligned}$$

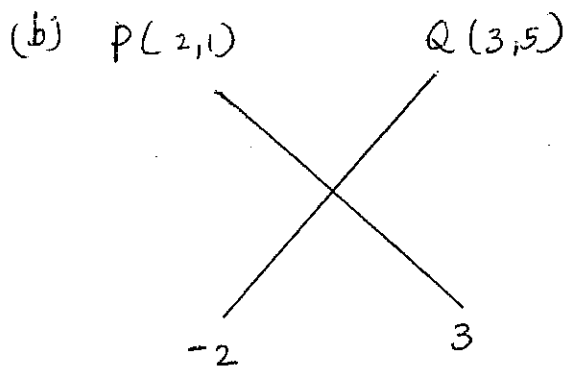
Question 4 (12 marks)

$$\begin{aligned} (a) & (5 - 2x^3)(1 + 2x)^5 \\ &= (5 - 2x^3)(1 + {}^5C_1(2x) + {}^5C_2(2x)^2 + {}^5C_3(2x)^3 + {}^5C_4(2x)^4 + {}^5C_5(2x)^5) \\ &= (5 - 2x^3)(1 + 10x + 40x^2 + 80x^3 + 80x^4 + 32x^5) \end{aligned}$$

Terms containing x^4 are

$$80 \times 5x^4 - 2 \times 10x^4 \quad (3)$$

$$\begin{aligned} \text{Coefficient of } x^4 &= 400 - 20 \\ &= \underline{\underline{380}} \end{aligned}$$



(4)

$$x = \frac{-2 \times 3 + 3 \times 2}{-2 + 3} = 0$$

$\therefore$ the point is $(0, -7)$

$$y = \frac{-2 \times 5 + 3 \times 1}{-2 + 3} = -7$$

(c) step 1 Testing $n=1$

when $n=1$, $LHS = 1(1+1)^2$
 $= 1 \times 4 = 4$

$$RHS = \frac{1 \times 2 \times 3 \times 8}{12} = 4$$

$LHS = RHS \therefore$ the result is true for $n=1$

step 2 Assume the result is true for $n=k$

$$\text{ie } 1 \times 2^2 + 2 \times 3^2 + 3 \times 4^2 + \dots + k(k+1)^2 = \frac{k(k+1)(k+2)(3k+5)}{12} \quad \text{--- (1)}$$

To prove that the result is true for $n=k+1$.

ie to prove that

$$1 \times 2^2 + 2 \times 3^2 + 3 \times 4^2 + \dots + k(k+1)^2 + (k+1)(k+2)^2$$

$$= \frac{(k+1)(k+2)(k+3)(3(k+1)+5)}{12}$$

$$= \frac{(k+1)(k+2)(k+3)(3k+8)}{12}$$

Let $S_k = 1 \times 2^2 + 2 \times 3^2 + 3 \times 4^2 + \dots + k(k+1)^2$ (5)

$$S_{k+1} = S_k + (k+1)(k+2)^2$$

$$= \frac{k(k+1)(k+2)(3k+5)}{12} + (k+1)(k+2)^2 \text{ by assumption (1)}$$

$$= \frac{k(k+1)(k+2)(3k+5) + 12(k+1)(k+2)^2}{12}$$

$$= \frac{(k+1)(k+2) [k(3k+5) + 12(k+2)]}{12}$$

$$= \frac{(k+1)(k+2)(3k^2+17k+24)}{12} = \frac{(k+1)(k+2)(k+3)(3k+8)}{12}$$

Thus the result is true for $n=k+1$.

Hence by Mathematical Induction, the result is true for all positive integral values of n .

Question 5 (23 marks)

(a) (i) $\int \frac{x+4}{x^2+8x-5} dx$

$$= \int \frac{2(x+4)}{2(x^2+8x-5)} dx$$

$$= \frac{1}{2} \int \frac{2x+8}{x^2+8x-5} dx \quad (3)$$

$$= \frac{1}{2} \log_e (x^2+8x-5) + C$$

(ii) $\int \left(\frac{1}{2} \sin 2x - \cos x \right) dx$

$$= \frac{1}{2} \times -\frac{\cos 2x}{2} - \sin x + C \quad (3)$$

$$= -\frac{1}{4} \cos 2x - \sin x + C$$

(iii) $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \sin^2 x dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \left(\frac{1-\cos 2x}{2} \right) dx$

$$= \frac{1}{2} \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} (1 - \cos 2x) dx$$

$$= \frac{1}{2} \left[x - \frac{\sin 2x}{2} \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}}$$

$$= \frac{1}{2} \left\{ \left(\frac{\pi}{3} - \frac{\sin \frac{2\pi}{3}}{2} \right) - \left(\frac{\pi}{4} - \frac{\sin \frac{\pi}{2}}{2} \right) \right\}$$

$$= \frac{1}{2} \left\{ \left(\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right) - \left(\frac{\pi}{4} - \frac{1}{2} \right) \right\}$$

$$= \frac{\pi - 3\sqrt{3} + 6}{24} \quad (4)$$

(iv) $\int_0^{\frac{\pi}{6}} \sin x \cos x dx = \int_0^{\frac{\pi}{6}} \frac{\sin 2x}{2} dx$

$$= \frac{1}{2} \int_0^{\frac{\pi}{6}} \sin 2x dx = \frac{1}{2} \left[-\frac{\cos 2x}{2} \right]_0^{\frac{\pi}{6}}$$

$$= -\frac{1}{4} [\cos 2x]_0^{\frac{\pi}{6}}$$

$$= -\frac{1}{4} (\cos \frac{\pi}{3} - \cos 0)$$

$$= -\frac{1}{4} \left(\frac{1}{2} - 1 \right) = -\frac{1}{4} \times -\frac{1}{2}$$

$$= \frac{1}{8} \quad (4)$$

$$(b) V = \pi \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \tan^2 x \, dx$$

$$= \pi \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} (\sec^2 x - 1) \, dx$$

$$= \pi \left[\tan x - x \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}} \quad (4)$$

$$= \pi \left\{ \left(\tan \frac{\pi}{3} - \frac{\pi}{3} \right) - \left(\tan \frac{\pi}{4} - \frac{\pi}{4} \right) \right\}$$

$$= \pi \left(\sqrt{3} - \frac{\pi}{3} - 1 + \frac{\pi}{4} \right)$$

$$= \frac{\pi (12\sqrt{3} - 12 - \pi)}{12}$$

$$(c) (i) y = \cos^3 x$$

$$\frac{d}{dx} (\cos^3 x) = 3 \cos^2 x \times -\sin x$$

$$= -3 \cos^2 x \sin x \quad (2)$$

$$\cos^2 x \sin x = -\frac{1}{3} \frac{d}{dx} (\cos^3 x)$$

$$(ii) \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \cos^2 x \sin x \, dx$$

$$= -\frac{1}{3} \int_0^{\frac{\pi}{2}} \frac{d}{dx} (\cos^3 x) \, dx$$

$$= -\frac{1}{3} \left[\cos^3 x \right]_0^{\frac{\pi}{2}} \quad (3)$$

$$= -\frac{1}{3} (\cos^3 \frac{\pi}{2} - \cos^3 0)$$

$$= -\frac{1}{3} (0 - 1)$$

$$= \frac{1}{3}$$

Question 6 (20 marks)

$$(a) p = 0.96 \quad q = 0.04$$

$$(i) p(x=20) = {}^{20}C_{20} p^{20} \quad (2)$$

$$= (0.96)^{20} = 0.442$$

$$(ii) \text{Probability that exactly two bulbs are defective}$$

$$= p(x=18)$$

$$= {}^{20}C_{18} p^{18} q^2 \quad (2)$$

$$= {}^{20}C_{18} (0.96)^{18} (0.04)^2$$

$$= 0.146$$

$$(iii) \text{Probability that at least one bulb is defective}$$

$$= 1 - \text{Probability that all bulbs are not defective} \quad (2)$$

$$= 1 - (0.96)^{20} = 0.558$$

$$(b) (i) \sin 2x = -\frac{\sqrt{3}}{2}, \quad 0 \leq x \leq 360^\circ$$

$$\text{Let } u = 2x$$

$$\sin u = -\frac{\sqrt{3}}{2}, \quad 0 \leq u \leq 720^\circ$$

u is in the 3rd or 4th quadrant.

$$\text{acute } \angle u = 60^\circ$$

$$180 + u = 240 \quad ; \quad 360 - u = 300^\circ$$

$$u = 240^\circ, 300^\circ, 600^\circ, 660^\circ$$

$$x = \underline{120^\circ, 150^\circ, 300^\circ, 330^\circ} \quad (4)$$

$$(ii) 3\cos x + 4\sin x = 4 \quad \text{--- (1)}$$

Substitute $\sin x = \frac{2t}{1+t^2}$ and

$$\cos x = \frac{1-t^2}{1+t^2} \quad \text{in (1)}$$

$$3 \times \frac{(1-t^2)}{1+t^2} + \frac{4 \times 2t}{1+t^2} = 4$$

$$\frac{3-3t^2}{1+t^2} + \frac{8t}{1+t^2} = 4$$

$$3-3t^2+8t = 4+4t^2$$

$$7t^2 - 8t + 1 = 0$$

$$(t-1)(7t-1) = 0$$

$$t = 1 \quad \text{or} \quad t = \frac{1}{7}$$

$$\tan \frac{x}{2} = 1 \quad \text{or} \quad \tan \frac{x}{2} = \frac{1}{7}$$

$$\tan \frac{x}{2} = 1 \quad \text{(5)}$$

$$\text{Let } u = \frac{x}{2}$$

$$\tan u = 1, \quad 0 \leq u \leq 180^\circ$$

$$u = 45^\circ$$

$$\therefore x = 90^\circ$$

$$\tan \frac{x}{2} = \frac{1}{7}$$

$$\text{Let } u = \frac{x}{2},$$

$$\tan u = \frac{1}{7}, \quad 0 \leq u \leq 180^\circ$$

$$u = 8^\circ 8' \quad \therefore x = 16^\circ 16'$$

$$\text{check for } x = 180^\circ$$

$$3 \times -1 + 4 \times 0 = 4$$

$$-3 = 4 \quad \text{not true}$$

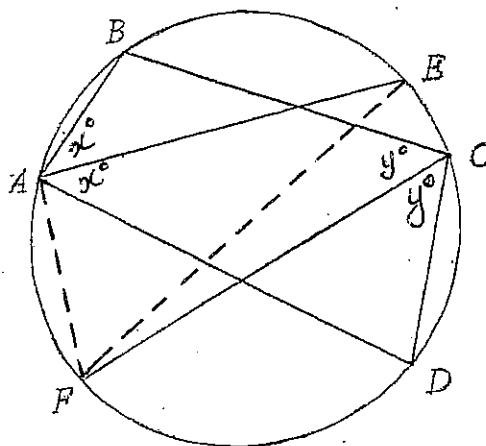
$$\therefore x = 180 \text{ is not a solution.}$$

The solutions are

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$$x = 16^\circ 16', 90^\circ$$

(c)



ABCD is a cyclic quadrilateral.

$x + x + y + y = 180$ (opposite angles of a cyclic quadrilateral are supplementary.)

$$2x + 2y = 180$$

$$x + y = 90$$

$\angle FAD = y$ (angles standing on the minor arc FD)

$$\angle FAE = \angle FAD + \angle EAD \quad \text{(5)}$$

$$= y + x$$

$$= 90^\circ$$

$\therefore EF$ is a diameter.

